

Лабораторная работа №03

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Цель работы

Приобретение практических навыков администрирования сетевых подсистем

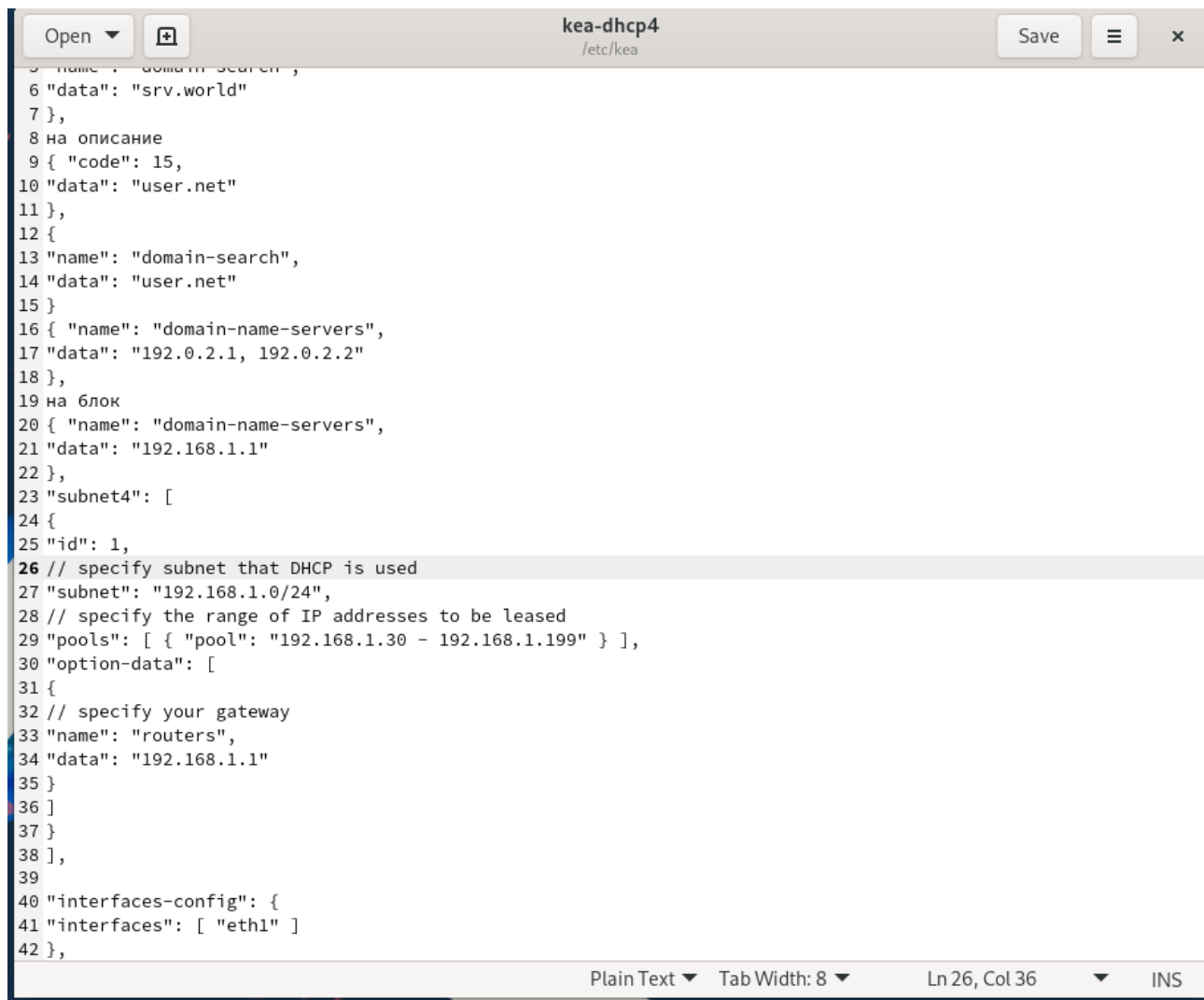
Загрузим нашу операционную систему и перейдём в рабочий каталог с проектом. Далее запустим виртуальную машину server (рис. @fig-1):

```
root@server:~
[vagrant@server ~]$ sudo -i
[root@server ~]# dnf -y install kea
Last metadata expiration check: 0:16:46 ago on Wed 04 Mar 2026 03:29:26 PM UTC.
Dependencies resolved.
=====
Package                               Arch      Version      Repository    Size
=====
Installing:
kea                                   x86_64    2.6.4-1.el9   epel          1.3 M
Installing dependencies:
kea-libs                             x86_64    2.6.4-1.el9   epel          3.0 M
log4cplus                           x86_64    2.0.5-15.el9  epel          337 k
mariadb-connector-c                  x86_64    3.2.6-1.el9_0 appstream     195 k
mariadb-connector-c-config            noarch    3.2.6-1.el9_0 appstream      9.8 k
postgresql-private-libs               x86_64    13.23-1.el9_7 appstream     140 k
=====
Transaction Summary
=====
Install 6 Packages

Total download size: 5.0 M
Installed size: 18 M
Downloading Packages:
(1/6): log4cplus-2.0.5-15.el9.x86_64.rpm                1.8 MB/s | 337 kB    00:00
(2/6): kea-libs-2.6.4-1.el9.x86_64.rpm                  4.9 MB/s | 3.0 MB    00:00
(3/6): mariadb-connector-c-3.2.6-1.el9_0.x86_64.rpm      443 kB/s | 195 kB    00:00
(4/6): kea-2.6.4-1.el9.x86_64.rpm                      1.9 MB/s | 1.3 MB    00:00
(5/6): mariadb-connector-c-config-3.2.6-1.el9_0.noarch.rpm 59 kB/s | 9.8 kB     00:00
(6/6): postgresql-private-libs-13.23-1.el9_7.x86_64.rpm 868 kB/s | 140 kB    00:00
-----
Total                                                    42 kB/s | 5.0 MB    02:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing    : postgresql-private-libs-13.23-1.el9_7.x86_64 1/6
  Installing    : mariadb-connector-c-config-3.2.6-1.el9_0.noarch 2/6
  Installing    : mariadb-connector-c-3.2.6-1.el9_0.x86_64 3/6
```

Рис. 1: Этап 1

На виртуальной машине server войдём под нашим пользователем и откроем терминал. Перейдём в режим суперпользователя и установим dhcp-server (рис. @fig-2):



```
Open [v] [x] kea-dhcp4 /etc/kea Save [≡] [x]
5 name: "domain-search",
6 "data": "srv.world"
7 },
8 на описание
9 { "code": 15,
10 "data": "user.net"
11 },
12 {
13 "name": "domain-name-servers",
14 "data": "user.net"
15 }
16 { "name": "domain-name-servers",
17 "data": "192.0.2.1, 192.0.2.2"
18 },
19 на блок
20 { "name": "domain-name-servers",
21 "data": "192.168.1.1"
22 },
23 "subnet4": [
24 {
25 "id": 1,
26 // specify subnet that DHCP is used
27 "subnet": "192.168.1.0/24",
28 // specify the range of IP addresses to be leased
29 "pools": [ { "pool": "192.168.1.30 - 192.168.1.199" } ],
30 "option-data": [
31 {
32 // specify your gateway
33 "name": "routers",
34 "data": "192.168.1.1"
35 }
36 ]
37 }
38 ],
39
40 "interfaces-config": {
41 "interfaces": [ "eth1" ]
42 },
```

Рис. 2: Этап 2

Скопируем файл примера конфигурации DHCP `dhcpd.conf.example` из каталога `/usr/share/doc/dhcp*` в каталог `/etc/dhcp` и переименуем его в файл с названием `dhcpd.conf` (рис. @fig-3):

Откроем файл `/etc/dhcp/dhcpd.conf` на редактирование. В этом файле заменим строки `option domain-name` и `option domain-name-servers`, раскомментируем строку `authoritative` и на базе одного из приведённых в файле примеров конфигурирования подсети зададим собственную конфигурацию DHCP-сети (рис. @fig-4):

Откроем файл `/etc/systemd/system/dhcpd.service` на редактирование и заменим в нём строку, указав интерфейс `eth1` (рис. @fig-5):

```

[ 44.621644] vmwgfx 0000:00:02.0: [drm] DMA map mode: Caching DMA mappings.
[ 44.624452] vmwgfx 0000:00:02.0: [drm] Legacy memory limits: URAM = 262144 kB
, FIFO = 2048 kB, surface = 262144 kB
[ 44.624462] vmwgfx 0000:00:02.0: [drm] MOB limits: max mob size = 131072 kB,
max mob pages = 262144
[ 44.624467] vmwgfx 0000:00:02.0: [drm] Max GMR ids is 8192
[ 44.624470] vmwgfx 0000:00:02.0: [drm] Max number of GMR pages is 1048576
[ 44.624472] vmwgfx 0000:00:02.0: [drm] Maximum display memory size is 262144
kB
[ 44.743257] vmwgfx 0000:00:02.0: [drm] Screen Target display unit initialized
[ 44.749736] vmwgfx 0000:00:02.0: [drm] Fifo max 0x00200000 min 0x00001000 cap
0x00000355
[ 44.750600] vmwgfx 0000:00:02.0: [drm] Using command buffers with DMA pool.
[ 44.750704] vmwgfx 0000:00:02.0: [drm] Available shader model: SM_5.
[ 44.754230] [drm] Initialized vmwgfx 2.20.0 20211206 for 0000:00:02.0 on mino
r 0
[ 44.770198] fbcon: vmwgfxdrmfb (fb0) is primary device
[ 44.780791] Console: switching to colour frame buffer device 160x50
[ 44.800855] vmwgfx 0000:00:02.0: [drm] fb0: vmwgfxdrmfb frame buffer device
[ 44.813079] snd_intel8x0 0000:00:05.0: intel8x0_measure_ac97_clock: measured 56364 usecs (2640 samples)
[ 44.813091] snd_intel8x0 0000:00:05.0: clocking to 49190
[ 50.307040] systemd-rc-local-generator[7391]: /etc/rc.d/rc.local is not marked executable, skipping.
[ 53.185485] NET: Registered PF_QIPCRTR protocol family
[ 58.765459] e1000: eth0 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: RX
[ 58.800401] IPv6: ADDRCONF(NETDEV_CHANGE): eth0: link becomes ready
[ 58.900760] e1000: eth1 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: RX
[ 58.965799] Warning: Unmaintained driver is detected: ip_set
[ 59.814644] IPv6: ADDRCONF(NETDEV_CHANGE): eth1: link becomes ready
[ 91.092609] block sda: the capability attribute has been deprecated.

```

Рис. 3: Этап 3

```

134 // When specifying options, you typically need to specify
135 // one of (name or code) and data. The full option specification
136 // covers name, code, space, csv-format and data.
137 // space defaults to "dhcp4" which is usually correct, unless you
138 // use encapsulate options. csv-format defaults to "true", so
139 // this is also correct, unless you want to specify the whole
140 // option value as long hex string. For example, to specify
141 // domain-name-servers you could do this:
142 // {
143 //     "name": "domain-name-servers",
144 //     "code": 15,
145 //     "csv-format": "true",
146 //     "space": "dhcp4",
147 //     "data": "user.net"
148 // }
149 // but it's a lot of writing, so it's easier to do this instead:
150 {
151     "name": "domain-name-servers",
152     "data": "192.168.1.1"
153 },
154
155 // Typically people prefer to refer to options by their names, so they
156 // don't need to remember the code names. However, some people like$
157 // to use numerical values. For example, option "domain-name" uses
158 // option code 15, so you can reference to it either by
159 // "name": "domain-name" or "code": 15.
160 {
161     "code": 15,
162     "data": "example.org"
163 },
164
165 // Domain search is also a popular option. It tells the client to
166 // attempt to resolve names within those specified domains. For
167 // example, name "foo" would be attempted to be resolved as
168 // foo.mydomain.example.com and if it fails, then as foo.example.com

```

Рис. 4: Этап 4

```

"subnet4": [
{
    // This defines the whole subnet. Kea will use this information to
    // determine where the clients are connected. This is the whole
    // subnet in your network.

    // Subnet identifier should be unique for each subnet.
    "id": 1,

    // This is mandatory parameter for each subnet.
    "subnet": "192.168.1.0/24",

    // Pools define the actual part of your subnet that is governed
    // by Kea. Technically this is optional parameter, but it's
    // almost always needed for DHCP to do its job. If you omit it,
    // clients won't be able to get addresses, unless there are
    // host reservations defined for them.
    "pools": [ { "pool": "192.168.1.30 - 192.168.1.199" } ],

    // These are options that are subnet specific. In most cases,
    // you need to define at least routers option, as without this
    // option your clients will not be able to reach their default
    // gateway and will not have Internet connectivity.
    "option-data": [
        {
            // For each IPv4 subnet you most likely need to specify at
            // least one router.
            "name": "routers",
            "data": "192.168.1.1"
        }
    ],

    // Kea offers host reservations mechanism. Kea supports reservations
    // by several different types of identifiers: hw-address
    // (hardware/MAC address of the client), duid (DUID inserted by the
    // client), client-id (client identifier inserted by the client) and
    // circuit-id (circuit identifier inserted by the relay agent).
    //

```

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Рис. 5: Этап 5

Остальные примеры задания конфигураций подсетей удалите.

3. Настройте привязку dhcpd к интерфейсу eth1 виртуальной машины server:

```

"interfaces-config": {
    "interfaces": [ "eth1" ]
},

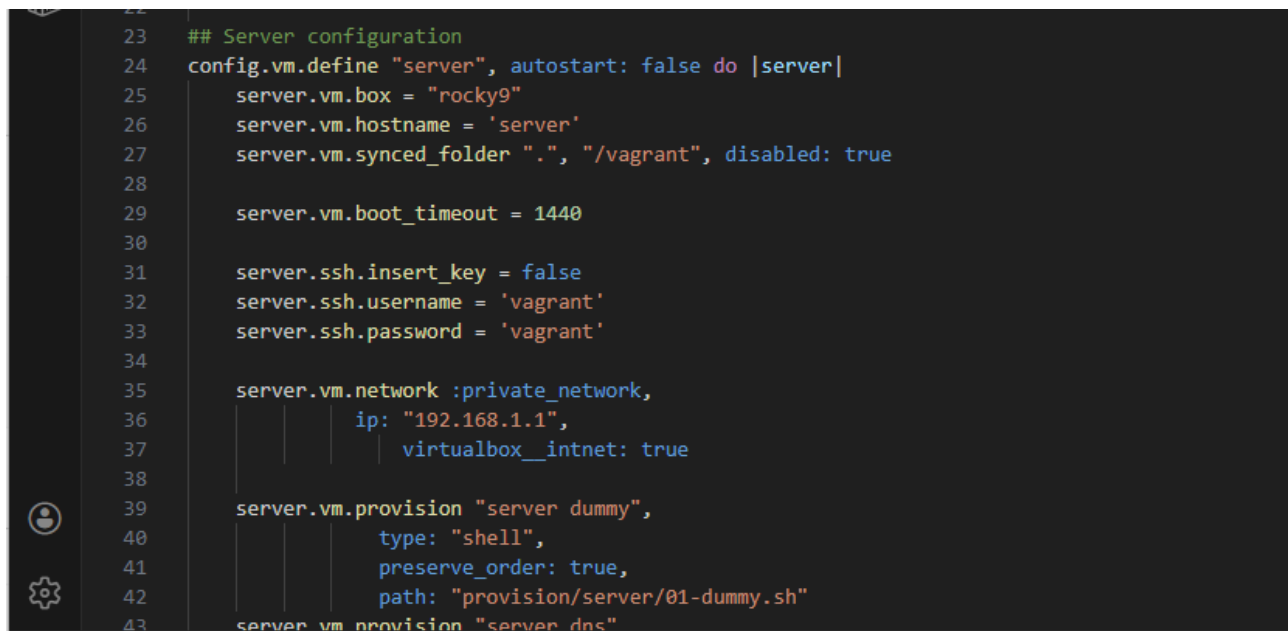
```

4. Проверьте правильность конфигурационного файла:
 kea-dhcp4 -t /etc/kea/kea-dhcp4.conf

Рис. 6: Этап 6

Перезагрузим конфигурацию systemd и разрешим загрузку DHCP-сервера при запуске виртуальной машины server (рис. @fig-6):

Добавим запись для DHCP-сервера в конце файла прямой DNS-зоны /var/named/master/ и в конце файла обратной зоны /var/named/master/rz/192.168.1. Перезапустим named и проверим, что можно обратиться к DHCP-серверу по имени (рис. @fig-7):



```
23  ## Server configuration
24  config.vm.define "server", autostart: false do |server|
25      server.vm.box = "rocky9"
26      server.vm.hostname = 'server'
27      server.vm.synced_folder ".", "/vagrant", disabled: true
28
29      server.vm.boot_timeout = 1440
30
31      server.ssh.insert_key = false
32      server.ssh.username = 'vagrant'
33      server.ssh.password = 'vagrant'
34
35      server.vm.network :private_network,
36          ip: "192.168.1.1",
37          virtualbox____intnet: true
38
39      server.vm.provision "server dummy",
40          type: "shell",
41          preserve_order: true,
42          path: "provision/server/01-dummy.sh"
43      server.vm.provision "server dns"
```

Рис. 7: Этап 7

Внесём изменения в настройки межсетевого экрана узла server, разрешив работу с DHCP, и восстановим контекст безопасности в SELinux (рис. @fig-8):

Этап выполнения работы №9

Этап выполнения работы №10

Этап выполнения работы №11

Этап выполнения работы №12

Этап выполнения работы №13

Этап выполнения работы №14

```

26 // DHCPv4 configuration starts here. This section will be read by DHCPv4 server
27 // and will be ignored by other components.
28 "Dhcp4": {
29     // Add names of your network interfaces to listen on.
30     "interfaces-config": {
31         // See section 8.2.4 for more details. You probably want to add just
32         // interface name (e.g. "eth0" or specific IPv4 address on that
33         // interface name (e.g. "eth0/192.0.2.1").
34         "interfaces": ["eth1" ]
35
36         // Kea DHCPv4 server by default listens using raw sockets. This ensures
37         // all packets, including those sent by directly connected clients
38         // that don't have IPv4 address yet, are received. However, if your
39         // traffic is always relayed, it is often better to use regular
40         // UDP sockets. If you want to do that, uncomment this line:
41         // "dhcp-socket-type": "udp"
42     },
43
44     // Kea supports control channel, which is a way to receive management
45     // commands while the server is running. This is a Unix domain socket that
46     // receives commands formatted in JSON, e.g. config-set (which sets new
47     // configuration), config-reload (which tells Kea to reload its
48     // configuration from file), statistic-get (to retrieve statistics) and many
49     // more. For detailed description, see Sections 8.8, 16 and 15.
50     "control-socket": {
51         "socket-type": "unix",
52         "socket-name": "kea4-ctrl-socket"
53     },
54
55     // Use Memfile lease database backend to store leases in a CSV file.
56     // Depending on how Kea was compiled, it may also support SQL databases

```

Рис. 8: Этап 8

```
[root@server ~]# kea-dhcp4 -t /etc/kea/kea-dhcp4.conf
2026-03-04 19:52:51.138 INFO [kea-dhcp4.hosts/27065.139698838464640] HOSTS_BACKENDS_REGISTER
ED the following host backend types are available: mysql postgresql
2026-03-04 19:52:51.144 WARN [kea-dhcp4.dhcp4srv/27065.139698838464640] DHCP4_SRV_MT_DISABLED_C
UEVE_CONTROL disabling dhcp queue control when multi-threading is enabled.
2026-03-04 19:52:51.145 WARN [kea-dhcp4.dhcp4/27065.139698838464640] DHCP4_RESERVATIONS_LOOK
UP_FIRST_ENABLED Multi-threading is enabled and host reservations lookup is always performed
first.
2026-03-04 19:52:51.146 INFO [kea-dhcp4.dhcp4srv/27065.139698838464640] DHCP4_SRV_CFGMGR_NEW_SU
BNET4 a new subnet has been added to configuration: 192.168.1.0/24 with params: t1=900, t2=18
00, valid-lifetime=3600
2026-03-04 19:52:51.146 ERROR [kea-dhcp4.dhcp4/27065.139698838464640] DHCP4_PARSER_FAIL faile
d to create or run parser for configuration element subnet4: specified reservation '192.0.2.2
01' is not within the IPv4 subnet '192.168.1.0/24'
Error encountered: specified reservation '192.0.2.201' is not within the IPv4 subnet '192.168
.1.0/24'
[root@server ~]# systemctl --system daemon-reload
[root@server ~]# systemctl enable kea-dhcp4.service
Created symlink /etc/systemd/system/multi-user.target.wants/kea-dhcp4.service → /usr/lib/syst
emd/system/kea-dhcp4.service.
[root@server ~]#
```

```

(gedit:27131): dconf-WARNING **: 19:54:00.846: failed to commit changes to dconf: Failed to e
xecute child process "dbus-launch" (No such file or directory)
[root@server ~]# /var/named/master/fz
-bash: /var/named/master/fz: No such file or directory
[root@server ~]# cd ..
[root@server /]# cd /var/named/master/fz
-bash: cd: /var/named/master/fz: No such file or directory
[root@server /]# cd /var
[root@server var]# cd master
-bash: cd: master: No such file or directory
[root@server var]# systemctl restart named
Failed to restart named.service: Unit named.service not found.
[root@server var]# ping dhcp.user.net
ping: dhcp.user.net: Name or service not known
[root@server var]# chown -R named:named /var/named
chown: invalid user: 'named:named'
[root@server var]#

```

Рис. 10: Этап 10

```

root@server:/var
cockpit dhcpv6-client imap imaps pop3 pop3s smtp ssh
[root@server var]# firewall-cmd --get-services
firewall-cmd --add-service=dhcp
firewall-cmd --add-service=dhcp --permanent
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd a
udit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp
bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ce
ph-mon cfengine checkmk-agent cockpit collectd condor-collector cratedb ctdb dds dds-multicas
t dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls docker-registry docker-swarm
dropbox-lansync elasticsearch etcd-client etcd-server finger foreman foreman-proxy freeipa-4
freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client gangli
a-master git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp ip
p-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kp
asswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-
controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-
scheduler-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvi
rt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memca
che minidlina mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula netbios-n
s netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storageconsol
e ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy promethe
us prometheus-node-exporter proxy-dhcp ps2link ps3netshr ptp pulseaudio puppetmaster quassel
radius rdp redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-clie
nt samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap s
pideroak-lansync spotify-sync squid sssd ssh steam-streaming svdrp svn syncthing syncthing-gu
i syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transm
ission-client upnp-client vdsms vnc-server warpinator wbm-http wbm-https wireguard ws-discov
ery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-c
lient xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
success
success
[root@server var]#

```

Рис. 11: Этап 11


```

us prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel
radius rdp redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-clie
nt samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap s
pideroak-lansync spotify-sync squid ssdp ssh steam-streaming svdrp svn syncthing syncthing-gu
i syncthing-relay synergy syslog syslog-tls telnet tentacle tftp tile38 tinc tor-socks transm
ission-client upnp-client vdsm vnc-server warpinator wbem-http wbem-https wireguard ws-discov
ery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-c
lient xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
success
success
[root@server var]# estorecon -vR /etc
bash: estorecon: command not found...
[root@server var]# restorecon -vR /etc
[root@server var]# restorecon -vR /var/named
restorecon: lstat(/var/named) failed: No such file or directory
[root@server var]# restorecon -vR /var/lib/kea/
[root@server var]# tail -f /var/log/messages
Mar  4 19:53:10 localhost systemd-rc-local-generator: /etc/rc.d/rc.local is not marked execut
able, skipping.
Mar  4 19:53:17 localhost systemd[1]: Reloading.
Mar  4 19:53:17 localhost systemd-rc-local-generator: /etc/rc.d/rc.local is not marked execut
able, skipping.
Mar  4 19:53:57 localhost gnome-shell[24477]: Window manager warning: Buggy client sent a _NE
T_ACTIVE_WINDOW message with a timestamp of 0 for 0x30000f8
Mar  4 19:53:57 localhost systemd[1]: Starting dnf makecache...
Mar  4 19:53:57 localhost dnf[27149]: Metadata cache refreshed recently.
Mar  4 19:53:58 localhost systemd[1]: dnf-makecache.service: Deactivated successfully.
Mar  4 19:53:58 localhost systemd[1]: Finished dnf makecache.
Mar  4 19:58:34 localhost systemd[1]: Starting PackageKit Daemon...
Mar  4 19:58:34 localhost systemd[1]: Started PackageKit Daemon.

```

Рис. 12: Этап 12



```

*01-routing.sh
1 echo "Provisioning script $0"
2 nmcli connection modify "eth1" ipv4.gateway "192.168.1.1"
3 nmcli connection up "eth1"
4 nmcli connection modify eth0 ipv4.never-default true
5 nmcli connection modify eth0 ipv6.never-default true
6 nmcli connection down eth0
7 nmcli connection up eth0

```

Рис. 13: Этап 13


```

xecute child process "dbus-launch" (No such file or directory)
[root@server /]# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe69:a8d prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:a00:27ff:fe69:a8d prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:69:0a:8d txqueuelen 1000 (Ethernet)
    RX packets 1479236 bytes 2019630895 (1.8 GiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 178954 bytes 12543603 (11.9 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.1 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::a00:27ff:fee3:341f prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:e3:34:1f txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 44 bytes 5322 (5.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 17 bytes 2039 (1.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 17 bytes 2039 (1.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[root@server /]#

```

Рис. 14: Этап 14

Этап выполнения работы №15

```

[root@server /]# cat /var/lib/kea/kea-leases4.csv
[root@server /]# mkdir -p /etc/named/keys
[root@server /]# mkdir -p /etc/named/keys
tsig-keygen -a HMAC-SHA512 DHCP_UPDATER > /etc/named/keys/dhcp_updater.key
[root@server /]# gedit etc/named/keys/dhcp_updater.key

(gedit:27344): dconf-WARNING **: 20:12:32.361: failed to commit changes to dconf: Fa

```

Рис. 15: Этап 15

Этап выполнения работы №16

Этап выполнения работы №17

Этап выполнения работы №18

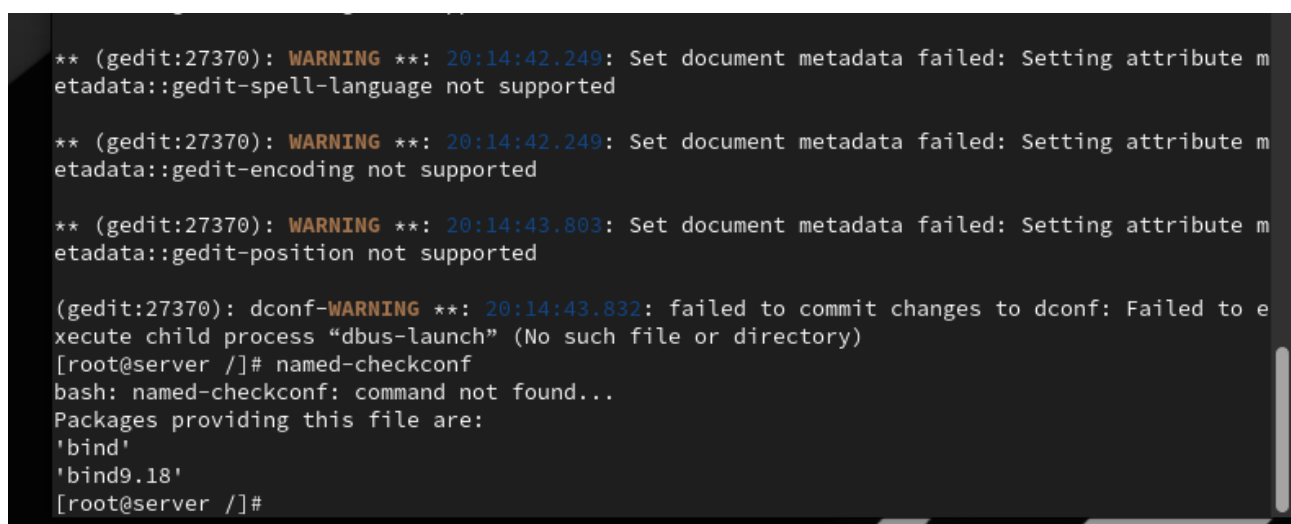
Этап выполнения работы №19

Этап выполнения работы №20



```
1 key "DHCP_UPDATER" {
2     algorithm hmac-sha512;
3     secret "gy3Cy8Pe3eEsw7TmjEjJf1/K7RzFMSn5bw2HZ0XMhu7NUQNd1017KDNi2ye5ZKJlpytiWY15jKv/hwvi7ig0rg==";
4 };
```

Рис. 16: Этап 16



```
** (gedit:27370): WARNING **: 20:14:42.249: Set document metadata failed: Setting attribute metadata::gedit-spell-language not supported
** (gedit:27370): WARNING **: 20:14:42.249: Set document metadata failed: Setting attribute metadata::gedit-encoding not supported
** (gedit:27370): WARNING **: 20:14:43.803: Set document metadata failed: Setting attribute metadata::gedit-position not supported

(gedit:27370): dconf-WARNING **: 20:14:43.832: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
[root@server /]# named-checkconf
bash: named-checkconf: command not found...
Packages providing this file are:
'bind'
'bind9.18'
[root@server /]#
```

Рис. 17: Этап 17



```
1 "tsig-keys" [
2 {
3   "name": "DHCP_UPDATER",
4   "algorithm": "hmac-sha512",
5   "secret":
6   "psFFSGdUIwK36l1G82L0w15PuIH4W5uB8h/cw7F0XDsjniBbwm/59tM+V3ydcCs15VLpe2pxlUlg↵
7 }
8 ]
```

Рис. 18: Этап 18

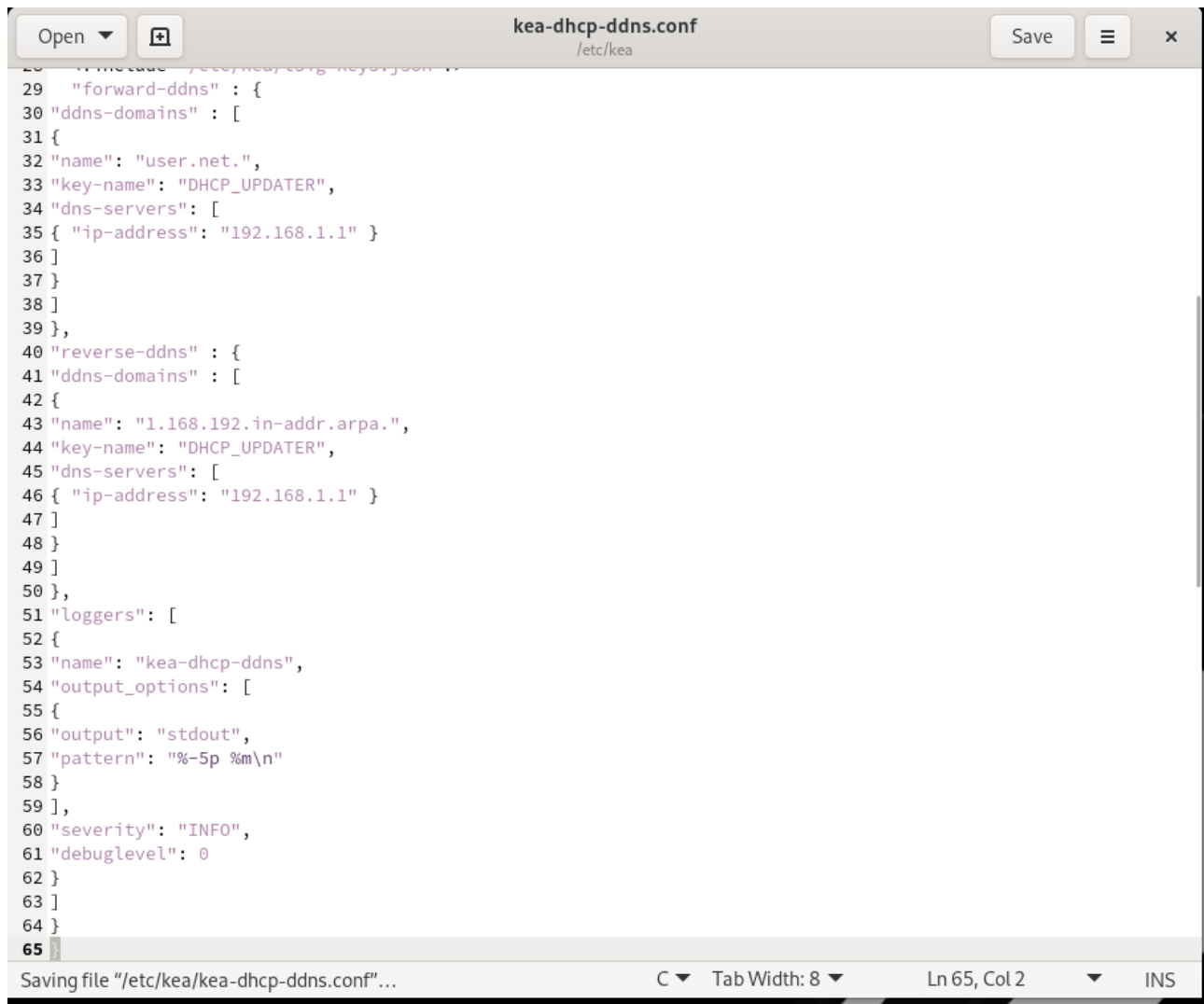


Рис. 19: Этап 19

```

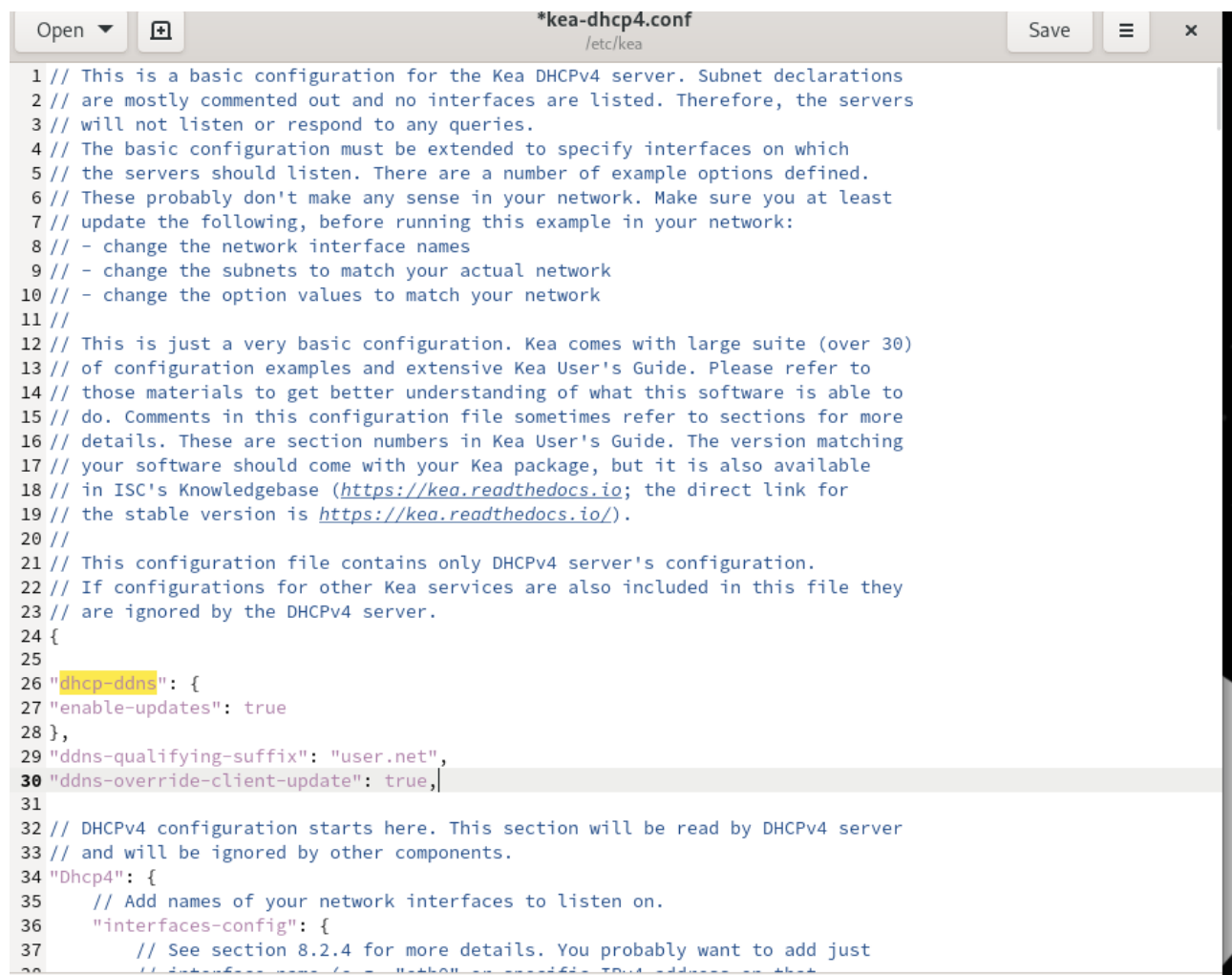
(gedit:27448): dconf-WARNING **: 20:18:21.001: failed to commit changes to dconf: Failed to e
xecute child process "dbus-launch" (No such file or directory)
[root@server ~]# chown kea:kea /etc/kea/kea-dhcp-ddns.conf
[root@server ~]# kea-dhcp-ddns -t /etc/kea/kea-dhcp-ddns.conf
Syntax check failed with: /etc/kea/tsig-keys.json:6.1: Invalid character: "
[root@server ~]# systemctl enable --now kea-dhcp-ddns.service
Created symlink /etc/systemd/system/multi-user.target.wants/kea-dhcp-ddns.service → /usr/lib/
systemd/system/kea-dhcp-ddns.service.
[root@server ~]# systemctl status kea-dhcp-ddns.service
* kea-dhcp-ddns.service - Kea DHCP-DDNS Server
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp-ddns.service; enabled; preset: disable>
   Active: failed (Result: exit-code) since Wed 2026-03-04 20:18:53 UTC; 14s ago
   Duration: 58ms
   Docs: man:kea-dhcp-ddns(8)
   Process: 27518 ExecStart=/usr/sbin/kea-dhcp-ddns -c /etc/kea/kea-dhcp-ddns.conf (code=ex>
   Main PID: 27518 (code=exited, status=1/FAILURE)
   CPU: 31ms

Mar 04 20:18:53 server.user.net systemd[1]: Started Kea DHCP-DDNS Server.
Mar 04 20:18:53 server.user.net kea-dhcp-ddns[27518]: 2026-03-04 20:18:53.091 INFO [kea-dhc>
Mar 04 20:18:53 server.user.net kea-dhcp-ddns[27518]: 2026-03-04 20:18:53.095 FATAL [kea-dhc>
Mar 04 20:18:53 server.user.net kea-dhcp-ddns[27518]: Service failed: Could Not load configu>
Mar 04 20:18:53 server.user.net systemd[1]: kea-dhcp-ddns.service: Main process exited, code>
Mar 04 20:18:53 server.user.net systemd[1]: kea-dhcp-ddns.service: Failed with result 'exit->
lines 1-15/15 (END)

```

Рис. 20: Этап 20

Этап выполнения работы №21



```
*kea-dhcp4.conf
/etc/kea

1 // This is a basic configuration for the Kea DHCPv4 server. Subnet declarations
2 // are mostly commented out and no interfaces are listed. Therefore, the servers
3 // will not listen or respond to any queries.
4 // The basic configuration must be extended to specify interfaces on which
5 // the servers should listen. There are a number of example options defined.
6 // These probably don't make any sense in your network. Make sure you at least
7 // update the following, before running this example in your network:
8 // - change the network interface names
9 // - change the subnets to match your actual network
10 // - change the option values to match your network
11 //
12 // This is just a very basic configuration. Kea comes with large suite (over 30)
13 // of configuration examples and extensive Kea User's Guide. Please refer to
14 // those materials to get better understanding of what this software is able to
15 // do. Comments in this configuration file sometimes refer to sections for more
16 // details. These are section numbers in Kea User's Guide. The version matching
17 // your software should come with your Kea package, but it is also available
18 // in ISC's Knowledgebase (https://kea.readthedocs.io; the direct link for
19 // the stable version is https://kea.readthedocs.io/).
20 //
21 // This configuration file contains only DHCPv4 server's configuration.
22 // If configurations for other Kea services are also included in this file they
23 // are ignored by the DHCPv4 server.
24 {
25
26     "dhcp-ddns": {
27         "enable-updates": true
28     },
29     "ddns-qualifying-suffix": "user.net",
30     "ddns-override-client-update": true,
31
32 // DHCPv4 configuration starts here. This section will be read by DHCPv4 server
33 // and will be ignored by other components.
34 "Dhcp4": {
35     // Add names of your network interfaces to listen on.
36     "interfaces-config": {
37         // See section 8.2.4 for more details. You probably want to add just
38         // interface name (e.g. "eth0") or specific IP address or that
```

Рис. 21: Этап 21

Этап выполнения работы №22

Выводы

В ходе выполнения лабораторной работы были приобретены практические навыки

Спасибо за внимание!

Вопросы?

```
root@server:~  
root@server:/var x root@server:/ x root@server:~ x  
tadata::gedit-spell-language not supported  
* (gedit:27584): WARNING **: 20:20:34.091: Set document metadata failed: Setting attribute m  
tadata::gedit-encoding not supported  
* (gedit:27584): WARNING **: 20:20:34.772: Set document metadata failed: Setting attribute m  
tadata::gedit-position not supported  
gedit:27584): dconf-WARNING **: 20:20:34.791: failed to commit changes to dconf: Failed to e  
ecute child process "dbus-launch" (No such file or directory)  
root@server ~]# kea-dhcp4 -t /etc/kea/kea-dhcp4.conf  
syntax check failed with: /etc/kea/kea-dhcp4.conf:26.1-11: syntax error, unexpected constant  
string, expecting Dhcp4  
root@server ~]# systemctl restart kea-dhcp4.service  
root@server ~]# systemctl status kea-dhcp4.service  
kea-dhcp4.service - Kea DHCPv4 Server  
Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4.service; enabled; preset: disabled)  
Active: failed (Result: exit-code) since Wed 2026-03-04 20:21:00 UTC; 8s ago  
Duration: 160ms  
Docs: man:kea-dhcp4(8)  
Process: 27628 ExecStart=/usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf (code=exited, st>  
Main PID: 27628 (code=exited, status=1/FAILURE)  
CPU: 61ms  
Mar 04 20:21:00 server.user.net systemd[1]: Started Kea DHCPv4 Server.  
Mar 04 20:21:00 server.user.net kea-dhcp4[27628]: 2026-03-04 20:21:00.871 INFO [kea-dhcp4.d>  
Mar 04 20:21:00 server.user.net kea-dhcp4[27628]: 2026-03-04 20:21:00.873 ERROR [kea-dhcp4.d>  
Mar 04 20:21:00 server.user.net kea-dhcp4[27628]: 2026-03-04 20:21:00.873 ERROR [kea-dhcp4.d>  
Mar 04 20:21:00 server.user.net systemd[1]: kea-dhcp4.service: Main process exited, code=exi>  
Mar 04 20:21:00 server.user.net systemd[1]: kea-dhcp4.service: Failed with result 'exit-code>  
lines 1-15/15 (END)
```

Рис. 22: Этап 22